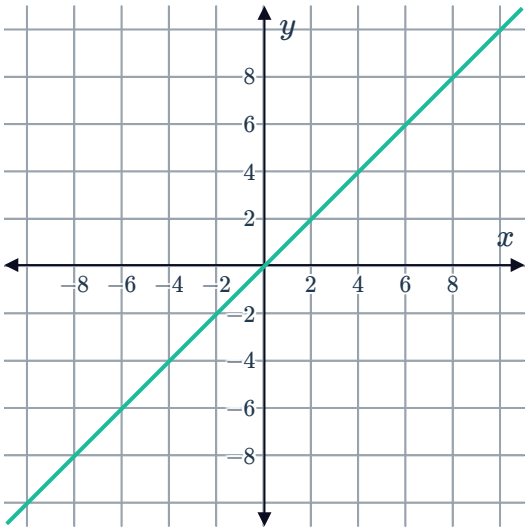
# Gradient function graphs

**1** Consider the graph of $y = x$:

**a** Find the gradient of the line at $x = 4$.

**b** Find the gradient at any value of $x$.

**c** What can be said about the gradient of a linear function?




**2** Consider the graph of $y = x^2$ and its gradient function $y = 2x$:

**a** What can be said of the sign of the gradient function when $y = x^2$ is increasing or decreasing?

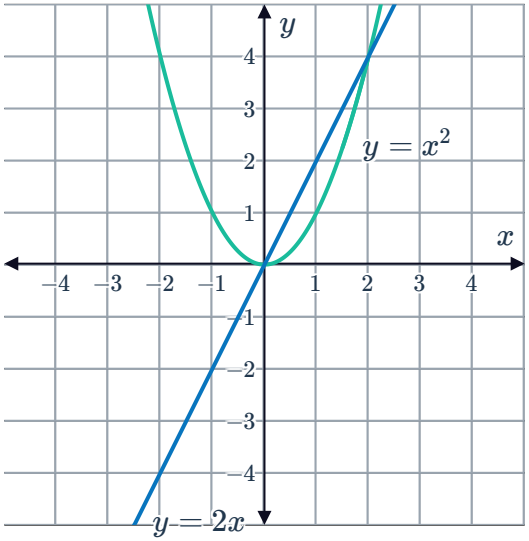
**b** For $x > 0$, is the gradient of the tangent positive or negative?

**c** For $x \geq 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**d** For $x < 0$, is the gradient of the tangent positive or negative?

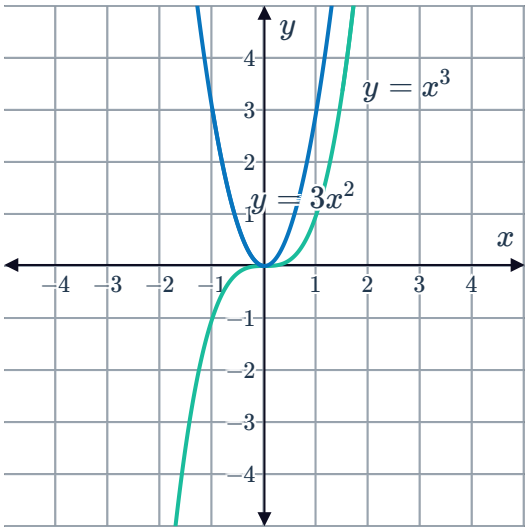
**e** For $x < 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**f** For $y = x^2$, the gradient of the tangent line changes at a constant rate. What type of function is the derivative of $y = x^2$?




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**3** Consider the graph of $y = x^3$ and its gradient function $y = 3x^2$.




**a** For $x > 0$, is the gradient of the tangent positive or negative?

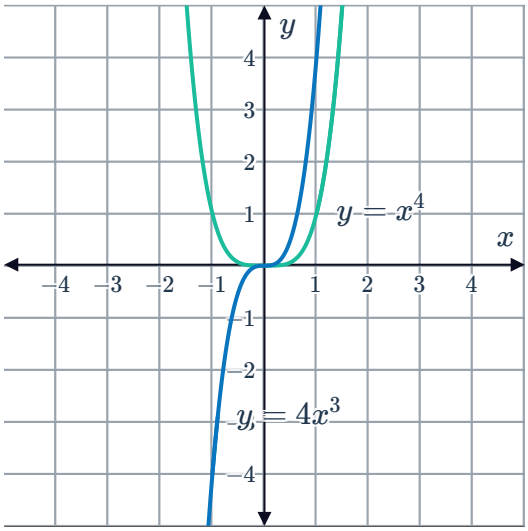
**b** For $x \geq 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**c** For $x < 0$, is the gradient of the tangent positive or negative?

**d** For $x < 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**e** For $y = x^3$, the gradient of the tangent line first decreases at a decreasing rate, then increases at an increasing rate. What type of function is the derivative of $y = x^3$?

**4** Consider the graph of $y = x^4$ and its gradient function $y = 4x^3$.




**a** For $x > 0$, is the gradient of the tangent positive or negative?

**b** For $x \geq 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**c** For $x < 0$, is the gradient of the tangent positive or negative?

**d** For $x < 0$, as the value of $x$ increases how does the gradient of the tangent line change?

**e** For $y = x^4$, the gradient of the tangent line is increasing, first at a decreasing rate and then at an increasing rate. What type of function is the derivative of $y = x^4$?

**5** Consider the functions $f(x) = x^5$ and $g(x) = x^4$.

**a** Sketch the graph of $f(x)$ and its derivative.

**b** Sketch the graph of $g(x)$ and its derivative.

**c** If the degree of a function is even, will the degree of its derivative function be odd or even?

**d** If the degree of a function is odd, will the degree of its derivative function be odd or even?

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## Algebraic differentiation

**6** Consider the function $f(x) = x^2$.

**a** Find the derivative of $f(x) = x^2$ from first principles.

**b** The derivative of $y = x^n$ for other values of $n$ can be found using first principles in a similar way. The results for $n = 2, 3, 4$ and $5$ are summarised in the following table:

| $y$ | $x^2$ | $x^3$ | $x^4$ | $x^5$ |
|---|---|---|---|---|
| $y'$ | $2x$ | $3x^2$ | $4x^3$ | $5x^4$ |

By observing the pattern in the table, find the derivative of the function $g(x) = x^8$.

**c** Write the general form of the derivative of $y = x^n$ for any value of $n$.

**7**

**a** For each of the following functions, state the degree of the gradient function:

**i** $y = x^2$ **ii** $y = x^3$ **iii** $y = x^4$

**b** Hence, state the degree of the derivative of a polynomial function of degree $n$.

**8** Find the derivative of the following functions with respect to $x$:

**a** $y = x^7$ **b** $y = x^9$ **c** $y = x^{-7}$ **d** $y = x^{-1}$

**e** $y = x^{\frac{6}{5}}$ **f** $y = x^{\frac{2}{3}}$ **g** $y = -6$ **h** $y = x^{0.6}$

**9** Consider the function $y = \sqrt{x}$.

**a** Rewrite the function in index form.

**b** Find the derivative of $y = \sqrt{x}$, in surd form.

**10** Consider the function $y = \dfrac{1}{x^2}$.

**a** Rewrite the function in negative index form.

**b** Find the derivative of $y = \dfrac{1}{x^2}$.

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**11** Consider the function $y = \frac{1}{\sqrt[4]{x}}$.

**a** Rewrite the function in negative index form.

**b** Find the derivative of $y = \frac{1}{\sqrt[4]{x}}$, in positive index form.

**12** Differentiate $y = t\sqrt{t}$.

**13** Differentiate $y = x^{-\frac{1}{5}}$ with respect to $x$. Express your answer in positive index form.

## Applications

**14** Find the gradient of $f(x) = x^4$ at $x = 2$. Denote this gradient by $f'(2)$.

**15** Consider the function $f(x) = x^9$.

**a** Find the gradient of the tangent to the function at $x = 1$.

**b** Find the equation of the tangent to the function at $x = 1$.

**16** Consider the function $f(x) = \frac{1}{\sqrt{x}}$.

**a** Find the gradient of the tangent to the function at $x = 1$.

**b** Find the equation of the tangent to the function at $x = 1$.

**17** David draws the graphs of $x^2$, $x^3$, $x^4$ and $x^5$ and draws the tangents to each one at the point where $x = 1$.

**a** David then notices where each of the tangents cut the $y$-axis and records this in the table below:

| Graph | $y$-intercept of the tangent line | Gradient of the tangent |
|---|---|---|
| $y = x^2$ | $(0, -1)$ | |
| $y = x^3$ | $(0, -2)$ | |
| $y = x^4$ | $(0, -3)$ | |
| $y = x^5$ | $(0, -4)$ | |

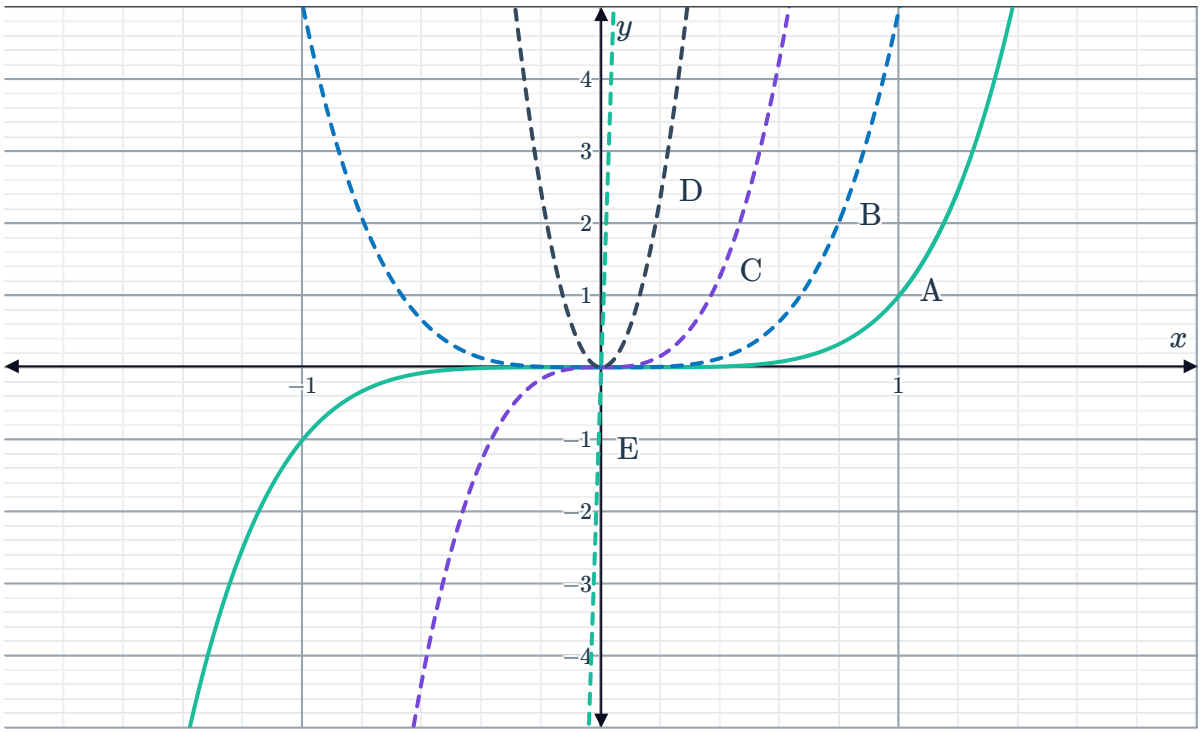
Complete the table by calculating the gradient of each of the tangents at $x = 1$.

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**b** Following the pattern in the table, what would be the gradient of the tangent to the graph of $y = x^n$ at the point where $x = 1$?

**c** Could the equation of the derivative of $y = x^6$ be $y' = x^5$? Explain your answer.

**18** The graph of $y = x^5$ is shown below labelled as $A$. Fiona then graphs the derivative of the function, labelling it as $B$. She then finds the derivative of graph $B$ to get graph $C$, then differentiates again to get graph $D$ and differentiates again to get graph $E$.




State whether the following statements are true about this sequence of derivatives:

**a** The derivative of a function is always positive when the function is negative, and negative when the function is positive.

**b** Each graph is a function of the form $ax^n$.

**c** For any value of $x$, the value of the derivative will always be greater than the value of the function.

**d** The degree of the derivative is always different to the degree of the function.

**19** Lucy used first principles to find the derivatives of $y = x^{\frac{1}{2}}$, $y = x^{\frac{3}{2}}$ and $y = x^{\frac{5}{2}}$. Her results are shown in the following table:

| $y$ | $x^{\frac{1}{2}}$ | $x^{\frac{3}{2}}$ | $x^{\frac{5}{2}}$ |
|---|---|---|---|
| $\frac{dy}{dx}$ | $\frac{1}{2}x^{-\frac{1}{2}}$ | $\frac{3}{2}x^{\frac{3}{2}}$ | $\frac{5}{2}x^{\frac{3}{2}}$ |

**a** Find the derivative of $y = x^{\frac{4}{3}}$.

**b** Find the derivative of $y = x^{\frac{3}{4}}$.

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